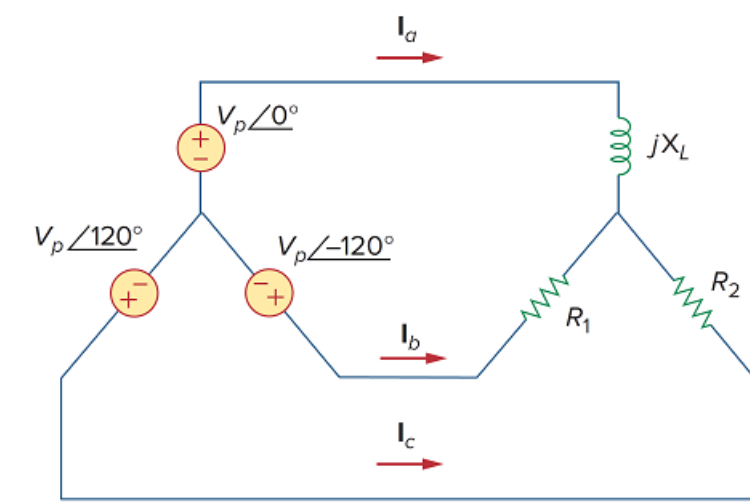


## Problem

Using Fig. 12.61, design a problem that will help other students better understand unbalanced three-phase systems.

Figure 12.61



## Step-by-step solution

## Step 1 of 1

Using Fig. 12.61, design a problem that will help other students better understand unbalanced three-phase systems.

Although there are many ways to solve this problem, this is an example based on the same kind of problem asked in the third edition.

## Problem

In the wye-wye system shown in Fig. 12.61, loads connected to the source are unbalanced.

(a) Calculate  $I_a$ ,  $I_b$ , and  $I_c$ . (b) Find the total power delivered to the load. Take  $V_P = 240$  V rms.

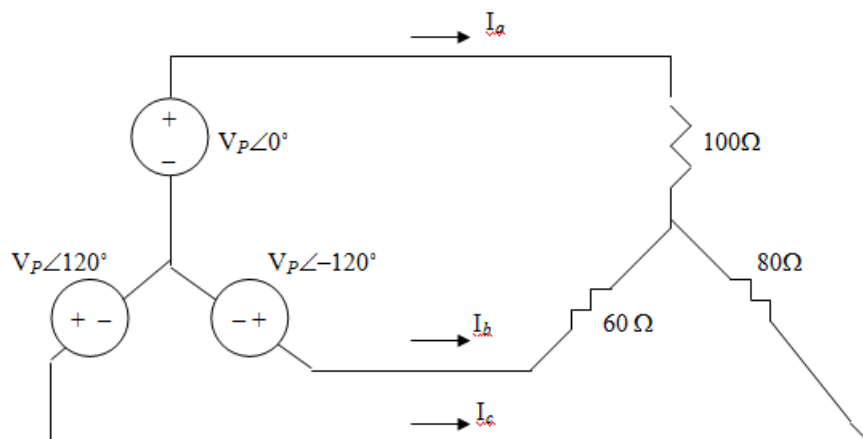
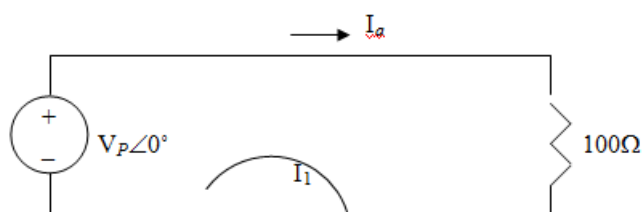


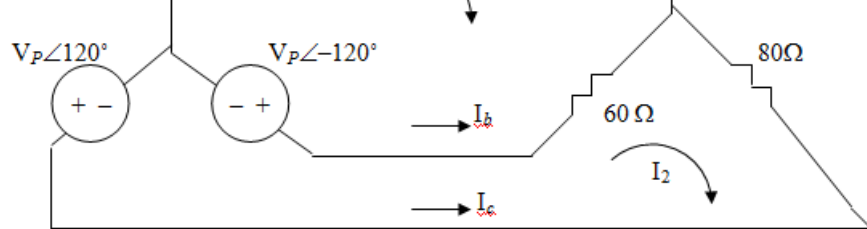
Figure 12.61

For Prob. 12.53.

## Solution

Applying mesh analysis as shown below, we get.





$$240 \angle -120^\circ - 240 + 160\mathbf{I_1} - 60\mathbf{I_2} = 0 \text{ or } 160\mathbf{I_1} - 60\mathbf{I_2} = 360 + j207.84 \quad (1)$$

$$240 \angle 120^\circ - 240 \angle -120^\circ - 60\mathbf{I_1} + 140\mathbf{I_2} = 0 \text{ or } -60\mathbf{I_1} + 140\mathbf{I_2} = -j415.7 \quad (2)$$

In matrix form, (1) and (2) become

$$\begin{bmatrix} 160 & -60 \\ -60 & 140 \end{bmatrix} \begin{bmatrix} \mathbf{I_1} \\ \mathbf{I_2} \end{bmatrix} = \begin{bmatrix} 360 + j207.84 \\ -j415.7 \end{bmatrix}$$

Using MATLAB, we get,

```
>> Z=[160,-60;-60,140]
```

Z =

```
160 -60
```

```
-60 140
```

```
>> V=[(360+207.8i);-415.7i]
```

V =

```
1.0e+002 *
```

```
3.6000 + 2.0780i
```

```
0 - 4.1570i
```

```
>> I=inv(Z)*V
```

I =

```
2.6809 + 0.2207i
```

```
1.1489 - 2.8747i
```

$\mathbf{I_1} = 2.681 + j0.2207$  and  $\mathbf{I_2} = 1.1489 - j2.875$

$\mathbf{I_a} = \mathbf{I_1} = \mathbf{2.69 \angle 4.71^\circ \text{ A}}$

$\mathbf{I_b} = \mathbf{I_2} - \mathbf{I_1} = -1.5321 - j3.096 = \mathbf{3.454 \angle -116.33^\circ \text{ A}}$

$\mathbf{I_c} = -\mathbf{I_2} = \mathbf{3.096 \angle 111.78^\circ \text{ A}}$

$$S_a = |\mathbf{I_a}|^2 Z_a = (2.69)^2 \times 100 = 723.61$$

$$S_b = |\mathbf{I_b}|^2 Z_b = (3.454)^2 \times 60 = 715.81$$

$$S_c = |\mathbf{I_c}|^2 Z_c = (3.0957)^2 \times 80 = 766.67$$

$$S = S_a + S_b + S_c = \underline{\underline{2.205 \text{ kVA}}}$$